# Raman Spectroscopy for Transparent Graphene Micro Electrode Arrays

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**ABSTRACT**

Raman spectroscopy is a widely used technique for characterizing carbon based nanomaterials such as carbon nanotubes and graphene. It is quick, non-destructive and accurate for determining the number of graphene layers. Here we present the assistance of this powerful technique in the fabrication of transparent graphene electrodes for micro electrode arrays (MEAs). MEAs are widely used to record the cellular activity from cells like neurons or cardiomyocytes. All typical standard electrode materials, Au, Pt, or TiN, are opaque, posing limitations for applications such as optogenetics and calcium imaging. Graphene offers the advantage of transparency, paving the way towards better simultaneous understanding of the cell-electrode interface and electrophysiology.

Detailed correlative scanning electron microscopy (SEM) and micro Raman spectroscopy is performed confirming the presence of graphene after each processing step. SEM is used for visualization of graphene and Raman provides information about the number of layers and contamination, if any. Figure 1 shows a SEM image of a graphene conduction line and electrode. Figure 1 (b) and (c) are the 2D and G peak maps, respectively, over the area seen in (a). Graphene microelectrodes of 30 micron diameter are fabricated with the assistance of Raman spectroscopy and successfully tested with cell cultures.

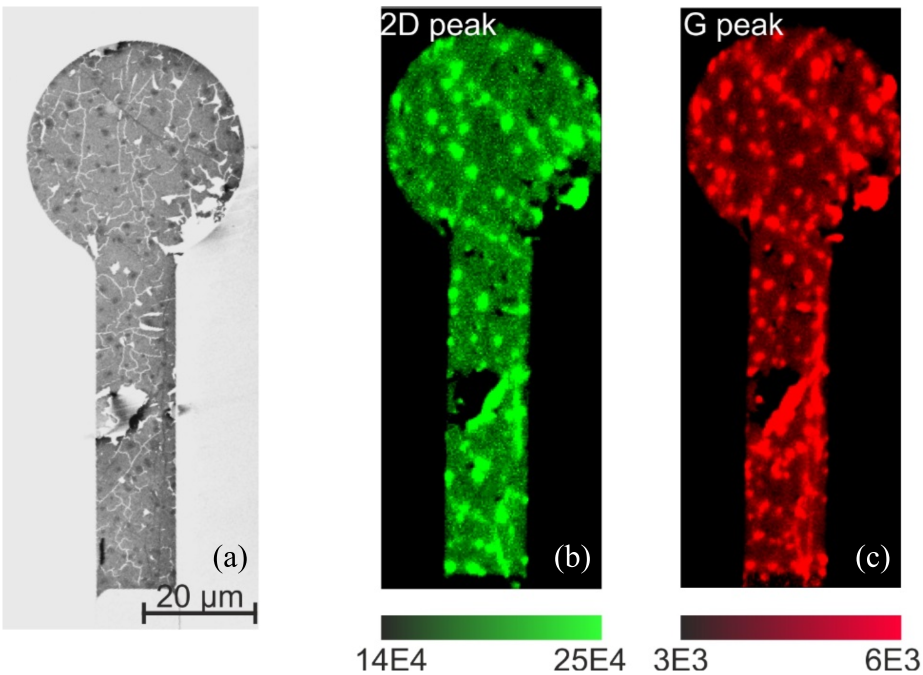


Figure 1: (a) Scanning electron micrograph of largely monolayer graphene conduction line and electrode. (b) 2D peak and (c) G peak Raman intensity maps, respectively.